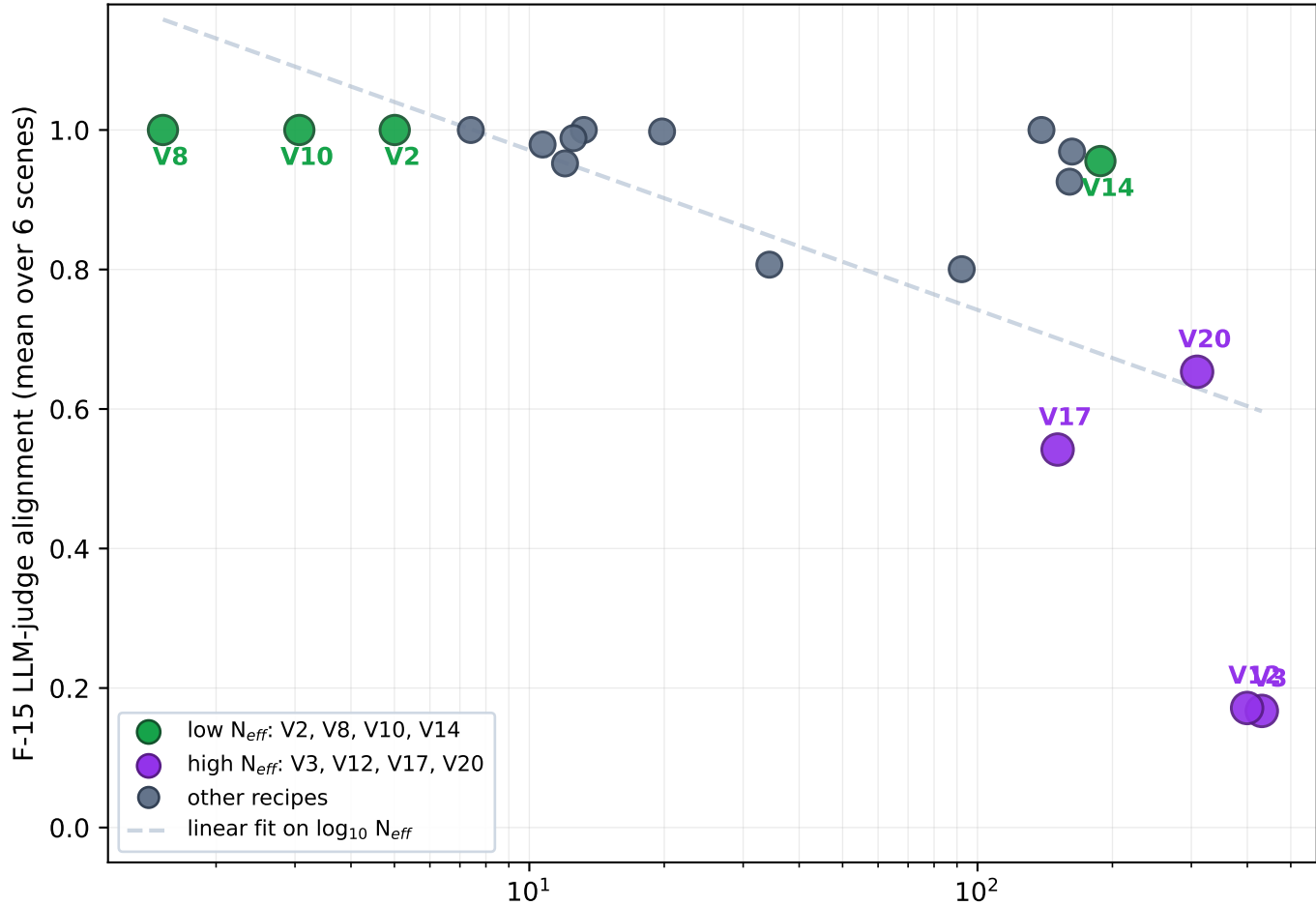


F-15 alignment against token-mass dispersion
 Spearman $\rho(\text{F-15}, N_{\text{eff}}) = -0.77$ (vs $\rho(\text{F-15}, |V|) = -0.68$; see companion figure)



Token-mass dispersion $N_{\text{eff}} = \exp H(\phi_k)$ (per-topic mean effective vocabulary, mean over 6 scenes,

$N_{\text{eff}} = \exp H(\phi_k)$ with ϕ_k the LDA topic-word distribution (mean over topics, then 6 scenes),
 from data/local/v_sweep/lda_fits/<scene>_<recipe>_uniform_Q8/phi.npy; F-15 from data/derived/v_sweep/f15_llm_alignment/.
 V2, V8 and V9 sit at F-15 = 1.0 by construction of the rule; V3 and V12 depend on the tie order of their
 equal-count documents (0.03 to 0.18).